Entropy Simulation

Project Idea

In this project, we simulate how the entropy of a material changes as its temperature increases. We use NumPy to perform calculations and Matplotlib to visualize the results.

1. Entropy Change Equation

For a material with constant specific heat capacity:

$$\Delta S = mc \ln\left(\frac{T_f}{T_i}\right)$$

This equation calculates the change in entropy as the material's temperature increases.

2. Heat Transfer Equation

The heat transferred to the material is calculated using:

$$Q = mc(T_f - T_i)$$

This equation calculates the amount of heat required to increase the material's temperature.

Variables

- **ΔS:** Change in entropy (J/K).
- **Q:** Heat transferred (J).
- **m:** Mass of the material (kg).
- **c:** Specific heat capacity (J/(kg·K)).
- **Ti:** Initial temperature (K).
- **Tf:** Final temperature (K).
- **ln:** Natural logarithm.

Libraries Used

- **NumPy:** Performs calculations for multiple temperature values.
- **Matplotlib:** Creates graphs showing the relationship between entropy, heat transfer, and temperature.

```
import numpy as np
import matplotlib.pyplot as plt

# Physical properties of the material
mass = 2.0              # Mass in kg
specific_heat = 900     # Specific heat capacity in J/(kg*K)
initial_temp = 300      # Initial temperature in Kelvin

# Generate final temperatures from 300 K to 1000 K
final_temp = np.linspace(300, 1000, 100)

# Calculate entropy change
entropy_change = mass * specific_heat * np.log(final_temp / initial_temp)

# Calculate heat transferred
heat_transferred = mass * specific_heat * (final_temp - initial_temp)

# Display selected results
print("Entropy Change Simulation")
print("-" * 35)

for i in [0, 25, 50, 75, 99]:
    print(
        f"Temperature: {final_temp[i]:.1f} K | "
        f"Heat: {heat_transferred[i]:.1f} J | "
        f"Entropy Change: {entropy_change[i]:.2f} J/K" )

# Create two plots
plt.figure(figsize=(12, 5))

# Plot 1: Entropy change versus temperature
plt.subplot(1, 2, 1)
plt.plot(final_temp, entropy_change, label="Entropy Change")
plt.xlabel("Final Temperature (K)")
```

```
    plt.ylabel("Entropy Change (J/K)")
    plt.title("Entropy vs Temperature")
    plt.grid(True)
    plt.legend()

    # Plot 2: Heat transferred versus temperature
    plt.subplot(1, 2, 2)
    plt.plot(final_temp, heat_transferred, label="Heat Transferred")
    plt.xlabel("Final Temperature (K)")
    plt.ylabel("Heat (J)")
    plt.title("Heat vs Temperature")
    plt.grid(True)
    plt.legend()

    # Adjust layout and display plots
    plt.tight_layout()
    plt.show()
```

```
Entropy Change Simulation
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Temperature: 300.0 K | Heat: 0.0 J | Entropy Change: 0.00 J/K
Temperature: 476.8 K | Heat: 318181.8 J | Entropy Change: 833.84 J/K
Temperature: 653.5 K | Heat: 636363.6 J | Entropy Change: 1401.51 J/K
Temperature: 830.3 K | Heat: 954545.5 J | Entropy Change: 1832.41 J/K
Temperature: 1000.0 K | Heat: 1260000.0 J | Entropy Change: 2167.15 J/K
```



